MEDICONNECT – HOSPITAL MANAGEMENT SYSTEM

Business Requirements Document (BRD) – Initial Draft

Document status: Working draft for learning and review

Project: MediConnect – Hospital Management System

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1. Introduction

This document records the business needs and initial system requirements for MediConnect, a proposed web-based Hospital Management System. It is based on the answers and observations given during a practice interview, together with the project charter. The aim is to explain the problems the hospital is trying to solve and the main things the new system should support.

This is an initial draft. The interview responses were practice assumptions rather than information collected from an actual hospital. The requirements should therefore be checked with real hospital management and users before they are treated as approved requirements.

2. Background and business problem

The hospital is assumed to rely mainly on paper-based records and manual processes, with information kept in separate files or departmental records. This can make everyday work slow, especially when many patients arrive at the same time.

- Patient registration can take time and create long queues.
- Medical records may be difficult to locate because files are not consistently organised. Patients may also have similar or identical names, which can make it harder to identify the correct record.
- Appointments may be handled manually using a first-come, first-served numbering process. Patients can wait longer when staff are also trying to retrieve paper files.
- Manual billing can lead to duplicate entries, typing mistakes, missed charges and incorrect calculations. This may result in some patients being overcharged or undercharged.
- Departments may work in silos, making it harder to share accurate and up-to-date patient information.
- During busy periods, the current process may not handle patient demand efficiently, reducing the number of patients the hospital can attend to.

3. Business objectives

1. Reduce delays during patient registration and improve the patient check-in process.
2. Make patient records easier for authorised staff to search, retrieve and update.
3. Improve coordination and information sharing between hospital departments.
4. Reduce manual billing errors and support accurate calculation and recording of charges and payments.
5. Improve appointment scheduling and reduce avoidable waiting time.
6. Give hospital management access to useful operational information through reports and a dashboard.
7. Provide patients with an appropriate way to view their own records and appointment information, subject to access and privacy rules.

4. Stakeholders and users

Stakeholder / user	Interest in the system	Main information needed
Hospital management	Operational performance and decision-making	Business goals, department needs, reports and approval of priorities
Receptionists	Registration and appointments	Patient details, patient search, check-in and appointment workflow
Doctors	Clinical information and patient care	Access to relevant patient history and clinical records
Nurses	Patient care and coordination	Timely access to relevant patient information and updates
Accountants / billing staff	Charges and payments	Accurate bills, payment records, corrections and reports
Patients	Their own care information	Access to permitted personal records and appointment details

5. Initial functional requirements

The requirements below translate the interview feedback into things the system should be able to do. They are proposed requirements and still need confirmation from the relevant users.

A. Patient registration and records

FR-01 Patient registration: The system shall allow authorised reception staff to create a patient profile and record the required patient details.

FR-02 Unique patient identification: The system shall assign each patient a unique patient ID so staff do not rely on names alone when identifying records.

FR-03 Patient search: The system shall allow authorised users to search for a patient using the patient ID and other approved search fields, such as name or phone number.

FR-04 Duplicate-record warning: The system shall warn staff when entered details appear to match an existing patient record, so possible duplicates can be checked before a new profile is created.

FR-05 Patient record retrieval: Authorised clinical staff shall be able to retrieve relevant patient records electronically without searching through physical files, subject to their assigned access permissions.

B. Appointment management

FR-06 Appointment management: The system shall allow authorised staff to create, view, reschedule and cancel appointments.

FR-07 Appointment status: The system shall show appointment status, such as scheduled, checked in, completed or cancelled.

C. Billing and payment management

FR-08 Billing: The system shall allow authorised billing staff to create and view patient bills based on recorded services and approved charges.

FR-09 Billing calculations: The system shall calculate bill totals from the entered charge items and applicable rates to reduce manual calculation errors.

FR-10 Payment records: The system shall record payments made against a bill and show the outstanding balance, where applicable.

FR-11 Billing correction and audit: The system shall allow authorised corrections to billing records while keeping a record of who made the change and when.

D. User access and information sharing

FR-12 Role-based access: The system shall provide access according to user roles, such as receptionist, doctor, nurse, accountant and administrator. Each role shall have defined permissions.

FR-13 Shared patient information: The system shall make current patient information available to authorised users across departments, subject to their assigned access permissions.

FR-14 Cross-department access: The system shall allow authorised clinical staff to access relevant patient records outside their own departments when required for patient care, subject to access restrictions and audit tracking.

FR-15 Patient portal: The system should provide patients with secure access to permitted personal information, including their appointment details and approved medical records.

E. Clinical record management and security

FR-16 Clinical record updates: The system shall allow authorised doctors to add consultation notes, diagnoses and treatment updates to patient records according to their permissions.

FR-17 Record bookmarking: The system should allow authorised doctors to bookmark patient records within the hospital platform for quicker future access. Patient records shall not be downloaded or saved as private copies under the current access policy.

FR-18 Clinical record correction history: The system shall preserve original clinical entries when corrections are made. Corrections shall be recorded as new entries, showing the person responsible and the date and time of the correction.

FR-19 Emergency access: The system shall allow designated department heads and directors to authorise emergency access to relevant patient information when normal access permissions are insufficient. Emergency access shall remain active for a maximum of one hour, provided no suspicious activity is detected. The system shall monitor emergency access activities, including unauthorised attempts to copy, transfer, download or capture patient information, as well as unusual access patterns. Where suspicious activity is detected, the system shall immediately suspend the emergency access, generate an alert and notify the appropriate security personnel and relevant management. All emergency access requests, approvals, activities and suspensions shall be recorded for audit and investigation. Fresh authorization is required if more time is needed.

F. Management and reporting

FR-20 Management dashboard: The system should provide management with summary information, such as registration activity, appointments and billing information.

FR-21 Reports: Authorised management users should be able to generate and view agreed operational reports.

6. Initial non-functional requirements

NFR-01 Security: Patient information must be protected from unauthorised access.

NFR-02 Privacy: Users should only view or change information needed for their assigned duties. Patient access should be limited to the patient's own permitted information.

NFR-03 Performance: Common activities such as patient search, record retrieval and appointment lookup should respond within an acceptable time under normal operating conditions. The exact target needs to be agreed and tested.

NFR-04 Availability: The system should be reliably available during agreed hospital operating hours.

NFR-05 Backup and recovery: Patient and billing information should be backed up regularly, with a recovery process in case of data loss or system failure.

NFR-06 Usability: The interface should be clear and straightforward for hospital staff with different levels of computer experience.

NFR-07 Data accuracy: The system should use validation checks to reduce incomplete, inconsistent or incorrectly entered information.

NFR-08 Auditability: Important activities, including record updates and billing corrections, should be traceable to the user who performed them.

NFR-09 Maintainability: The system should be designed so that authorised technical staff can maintain it and introduce approved improvements.

7. Business rules and points to confirm

- A patient should have one primary patient ID. Similar names must not be treated as proof that two records belong to the same person.
- Only authorised staff should create, view, update or correct patient information.
- A patient should only be able to access their own permitted information after identity verification.
- A bill should show the services or items charged, the amount for each item, the total, payments received and any outstanding balance.
- Any correction to a completed bill or patient record should be traceable and follow an agreed approval process.

These are sensible starting rules, not confirmed hospital policies. The hospital must decide who can approve billing adjustments, what information patients can see, how patient identity is verified and how long records are retained.

8. Scope notes

The initial MediConnect release is intended to cover patient registration and profiles, appointment management, electronic medical records, doctor and staff accounts, billing and payment records, an administrative dashboard, authentication and role-based access. The original project charter identifies telemedicine, ambulance tracking, advanced pharmacy inventory, full laboratory integration, insurance claims automation and a dedicated mobile app as outside the initial release. Those items should not be added without reviewing the scope, time, budget and approval.

9. Assumptions and limitations

The interview was a learning exercise. The comments about paper files, queues, manual appointments, billing errors and departmental silos were reasonable assumptions based on the case study, but they have not been verified with actual hospital staff. The example of a hospital serving 1,000 patients but managing 50 percent or less was an illustration raised during the discussion, not measured hospital data. It should not be used as a project performance baseline unless real figures are collected.

The system's exact performance targets, patient data fields, appointment rules, billing workflow, payment methods, report contents, integrations, data migration approach and patient portal permissions still need further discussion.

10. Questions for the next stakeholder interviews

Receptionists

8. What information do you collect when registering a new patient?
9. How do you check whether a patient already has a record?
10. How are appointments booked, changed and cancelled?
11. What happens when a patient's file cannot be found?
12. What would make registration and check-in easier?

Doctors

13. Which parts of a patient's medical history do you need to see during a consultation?
14. What information do you need to add after seeing a patient?
15. When should other staff be able to view clinical notes?
16. What problems do you face when patient records are incomplete or unavailable?
17. What would make the system useful during consultations?

Nurses

18. What patient information do you need during routine care?
19. How do you currently receive updates or instructions from doctors?
20. What information do you record during patient care?
21. What delays occur when you need a patient's file or history?
22. What access should nurses have in the system?

Accountants / billing staff

23. How are charges entered and bills prepared today?
24. How are payments recorded and receipts issued?
25. What causes incorrect, duplicate or missing charges?
26. Who is allowed to correct a bill, and how is the correction approved?
27. What billing reports does management need?

11. Recommended next steps

28. Review this draft with the project sponsor and confirm the business objectives and priorities.
29. Interview receptionists, doctors, nurses and billing staff to understand their actual workflows.
30. Record each stakeholder's needs and note where their requirements differ.
31. Map the current process and the proposed future process for registration, appointments, records and billing.
32. Agree the MVP requirements, priorities and acceptance criteria before the team begins detailed design and development.

12. Review and approval

Business representative: _____ Date: _____

Project Manager: _____ Date: _____

Business Analyst: _____ Date: _____

Status: Draft , not yet reviewed or approved by a real hospital sponsor.